

Measuring Common Ground in LLMs: From Category Norms to Tacit Coordination

XINYUE PENG, Peking University, China

We present two benchmarks for evaluating social cognition in large language models (LLMs). **Benchmark 1** replicates a classic human category-norm paradigm [?]: given a category label (e.g., “a fruit”), the model freely generates as many exemplars as possible. We compare the resulting exemplar frequency distributions against human normative data from US participants, measuring rank correlation and top- K overlap. **Benchmark 2** introduces a tacit coordination task [?]: two independent LLM agents see the same 4-option category but with options shuffled in different random orders, and must independently select the same option to earn a reward. This tests whether models converge on the same *focal point* via shared cultural salience, without any communication. Together, the two benchmarks span complementary levels of analysis—Benchmark 1 probes the structure of semantic knowledge itself, while Benchmark 2 tests whether that knowledge supports implicit coordination between agents.

CCS Concepts: • **Computing methodologies** → **Natural language processing**.

Additional Key Words and Phrases: social cognition, category norms, tacit coordination, focal points, common ground, large language models

ACM Reference Format:

Xinyue Peng. 2026. Measuring Common Ground in LLMs: From Category Norms to Tacit Coordination. *J. ACM* 37, 4, Article 111 (August 2026), 5 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

A core component of social cognition is *common ground*—the shared knowledge, beliefs, and assumptions that people expect others to hold [?]. When asked to name “a fruit,” most English speakers say “apple” first; when asked to independently name a US city for a coordination bonus, most say “New York.” Both phenomena reveal the same underlying capacity: shared cultural knowledge that makes certain items more *salient* than others.

We propose two benchmarks that probe this capacity in LLMs at different levels:

Benchmark 1 (Category-Norm Replication) directly replicates a well-established cognitive psychology paradigm. We adopt the 70-category semantic generation task from Castro, Curley, & Hertzog [?], in which participants listed as many exemplars as possible for each category label. We transfer this task to LLMs with minimal modification and compare model-generated exemplar distributions against the original human norms. This tests whether LLMs have internalized the same category structure as humans—not just which items belong to a category, but how *accessible* each item is.

Benchmark 2 (Tacit Coordination) goes further: it tests whether shared salience knowledge can support *coordination* between two independent agents. Drawing on Schelling’s [?] focal-point theory, we present two LLM agents with the same 4-option category but with options shuffled in

Author’s Contact Information: Xinyue Peng, Peking University, Beijing, China, xinyueshine@pku.edu.cn.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 1557-735X/2026/8-ART111

<https://doi.org/XXXXXXX.XXXXXXX>

different orders, and ask them to independently choose the same option. Success requires both agents to converge on the same culturally salient item without communication.

The two benchmarks form a natural progression: Benchmark 1 asks “does the model know what humans know?” and Benchmark 2 asks “can two models use that knowledge to coordinate?”

2 Related Work

Category Norms. Category norm studies collect the exemplars people generate for semantic categories, revealing frequency rankings and typicality gradients. The classic Battig & Montague [?] norms were updated by Castro et al. [?] with 70 categories and modern US participants. These norms are widely used as stimuli in memory and language research.

Focal Points and Coordination. Schelling [?] showed that people coordinate without communication by recognizing focal points—outcomes that “stand out” due to salience or convention [?]. Lewis [?] formalized this as a coordination game.

LLM Benchmarks. Existing benchmarks test knowledge (MMLU [?]), reasoning (BIG-bench [?]), or general ability (AGIEval [?]). No benchmark specifically tests whether LLMs replicate human category-norm structure or can use shared salience for tacit coordination. The Kaggle Measuring Progress Toward AGI competition [?] identifies social cognition as a key dimension.

3 Benchmark 1: Category-Norm Replication

3.1 Task

We adopt the 70 category labels from Castro, Curley, & Hertzog [?] verbatim (Table ??). For each category, the LLM is prompted:

List as many examples of “[category label]” as you can. Write one example per line.

This mirrors the original human task, where participants had 30 seconds to generate exemplars freely. We run each prompt $N = 100$ times (temperature 0.7) to build an exemplar frequency distribution, simulating 100 independent participants.

3.2 Categories

The 70 categories span a broad range of semantic domains. A subset is shown below; the full list is in the supplementary materials.

Table 1. Benchmark 1: Selection of the 70 category labels (Castro et al., 2021).

Domain	Sample Categories
Natural kinds	A bird; A fish; A flower; A fruit; A tree
Artifacts	A type of car; A type of fabric; A weapon
Social	A female first name; An occupation; A sport
Spatial	A city; A state; A country
Abstract	A color; A unit of time; A part of speech
Total	70 categories

3.3 Alignment Metrics

We compare LLM-generated exemplar distributions against the human normative data from [?] using four metrics:

(1) Top- K overlap. The Jaccard similarity between the top- K most frequent exemplars from humans and the LLM:

$$\text{TopK}(cat, K) = \frac{|H_K \cap L_K|}{|H_K \cup L_K|} \quad (1)$$

where H_K and L_K are the sets of top- K exemplars for humans and LLM respectively.

(2) Rank correlation. Spearman's ρ between human and LLM frequency rankings across exemplars appearing in either top-20 list.

(3) First-response match. Whether the LLM's most frequent exemplar matches the human modal response:

$$\text{FRM}(cat) = \mathbb{1}[\arg \max_L = \arg \max_H] \quad (2)$$

(4) Frequency correlation. Pearson's r between human and LLM frequency vectors over shared exemplars.

The overall Benchmark 1 score is the average across all 70 categories.

4 Benchmark 2: Tacit Coordination

4.1 Task

Two independent LLM agents play the following coordination game:

You are playing a coordination game with another agent. If you both select the SAME option, you each earn \$1. You cannot communicate. Note: The other agent sees the SAME options, but in a DIFFERENT order.

Each agent is shown a neutral category label and four options in a *randomized* order. Option order is shuffled independently for each agent, preventing trivial coordination via position bias.

4.2 Design Principles

(1) No leading language. Categories are strictly neutral—no “most famous,” “traditional,” “best,” or “typical.” Coordination depends solely on the agent's own salience judgment.

(2) Four options at the same semantic level. All four options are roughly equally plausible, yet one is culturally salient enough to serve as a focal point.

(3) Bilingual and cross-cultural. Every item exists in both English and Chinese. Culture-sensitive domains include separate US-context and China-context items.

4.3 Domain Taxonomy

We organize 31 domains into 8 categories by the *cognitive mechanism* supporting coordination (Table 1). Weakly culture-dependent domains (perceptual, embodied, functional) are balanced with strongly culture-dependent ones (holidays, brands, digital conventions), enabling analysis of cultural specificity effects.

Each domain contains 20 items (620 total, 1,240 bilingual).

4.4 Evaluation: Coordination Rate

For each item, we query the model 100 times with randomized option orders, simulating 100 agents. The pairwise coordination rate is:

$$\text{CoordRate}(item) = \frac{\sum_{i < j} \mathbb{1}[c_i = c_j]}{\binom{100}{2}} \quad (3)$$

Chance is 0.25 (random among 4); perfect coordination is 1.0.

Table 2. Benchmark 2: Domain taxonomy. 31 domains in 8 categories, organized by coordination mechanism.

Category	Domains (<i>n</i>)	Primary Cognitive Mechanism	Culture
Perception	Colors; Shapes; Spatial Directions; Extremes (4)	Perceptual prototype; symmetry bias; spatial default	Unspecific
Symbolism	Numbers; Time Anchors; Emotions (3)	Symbolic salience; roundness; temporal convention	Unspecific
Biology	Animals; Plants; Fruits; Body Parts; Senses (5)	Biological prototype; familiarity; embodied centrality	Unspecific
Artifacts	Tools; Clothing; Vehicles; Furniture (4)	Functional typicality; affordance salience	Unspecific
Places	Rooms; Public Places; Institutions; Geographic Entities (4)	Spatial routine; meeting-point scripts; collective prominence	Mixed*
Norms	Family Roles; Occupations; Social Norms (3)	Role prototype; prestige salience; normative scripts	Mixed*
Culture	Holidays; Food; Drinks; Famous People; Media; Brands (6)	Collective memory; ritual salience; media exposure	Specific
Digital	Digital Platforms; Internet Culture (2)	Platform familiarity; interface conventions	Specific
Total	31 domains		

*Mixed: most domains unspecific; Public Places, Geographic Entities, Social Norms are culture-specific.

5 Results

[XXX – XX benchmark XXXXXXXXXXXXX]

Benchmark 1 XXX.

- (1) XX Castro et al. (2021) XX normative data XXX JSON XX
- (2) XX python scripts/benchmark1_run.py --runs 100 XX 70 XXXX LLM XXXX
- (3) XX python scripts/benchmark1_analyze.py --human data/human_norms.json XX alignment
- (4) XX 5–6 XXXXXXXXXXXXX

Benchmark 2 XXX.

- (1) XX python scripts/generate_dataset.py --all-domains XX 620 items
- (2) XX python scripts/evaluate_dataset.py XXXXXX
- (3) XXXXXXXX item XX 100 XXXXXXXX
- (4) XX coordination rate XX cross-cultural XX

6 Discussion

[XXX – XXXXX]

- Benchmark 1: LLM XX category structure XXXXXXXXXXXXX category XXXXX
- Benchmark 2: XXXXXXXXXXXXXXXXXXXXX
- XX benchmark XXXX category norm alignment XXXXX coordination XXXXX
- XXXXXXXXXXXXXXXXXXXXX
- XXXXXXXX

7 Conclusion

[XXX – XXXXX]

- XX two-benchmark frameworkXX category norm replication XX tacit coordination

- J. ACM, Vol. 37, No. 4, Article 111. Publication date: August 2026.